

Financial Goal Planner: An AI-Driven Goal-Based Investment Planning and Risk-Aware Recommendation System

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Contents

Introduction

This project proposes a **mobile application** that recommends personalized savings plans and investment allocations across mutual funds, equity, and debt instruments based on a user's financial goal, horizon, and risk profile

The system aims to provide adaptive, risk-aware, and horizon-sensitive investment guidance through a mobile-first interface.

Problem Statement

Individuals planning for financial goals – such as retirement, a child's education, or purchasing a vehicle – face difficulty determining **how much to save, for how long, and where to invest**, because the appropriate level of investment risk depends heavily on the goal's remaining time horizon.

Objectives

- ① To design a mobile application that allows users to define financial goals with a target amount and time horizon.
- ② To build a recommendation engine that dynamically adjusts asset allocation across mutual funds, equity, and debt instruments.
- ③ To provide an intuitive mobile dashboard for tracking goal progress, required SIP contributions, and recommended portfolio adjustments over time.

Literature Summary (1/2)

Ref	Author(s)	Title	Journal	Year
1	Karzanov et al.	Regret-Optimized Portfolio Enhancement [BASE PAPER]	arXiv	2025
2	Dhashanamoorthi et al.	AI-driven robo-advisors on personal financial mgmt	Int. J. of Sci. and Research	2024
3	Sunilkumar	AI Revolution in Wealth Management	J. of Info Systems Eng.	2025
4	Ansari et al.	Multifaceted Approach to Stock Market Trading Using RL	IEEE Access	2024
5	Sattar et al.	RMS-Driven Deep RL for Optimized Portfolio Management	IEEE Access	2025
6	Yin & Guo	Complex forecasting via chain-of-thought of LLMs	Science Direct	2025
7	Benhamou et al.	Detecting and adapting to crisis pattern with context DRL	arXiv	2020
8	Samanta et al.	Effectiveness of SIP Investment Among Retail Investors	Int. J. for Novel Research	2024
9	Kanagaraj et al.	Potential of AI in personal financial planning	arXiv	2024
10	Boaretto & Medeiros	Forecasting inflation using disaggregates and ML	arXiv	2023
11	Solanki & Jha	GAF-ResNet-LSTM Framework for Asset Selection	IEEE Access	2025
12	Kevin & Yugopuspito	Hybrid LSTM and PPO Networks for Portfolio Optimization	arXiv	2025

Literature Summary (2/2)

Ref	Author(s)	Title	Journal	Year
13	Olamijuwon & Zouo	ML in budget forecasting for corporate finance	Open Access Research	2024
14	Spadea & Seneviratne	Parallel and Multi-Stage Knowledge Graph Retrieval	arXiv	2025
15	Patra et al.	AI-Driven Goal Based Financial Planning System	Rademics Research Inst.	2025
16	Lwele et al.	Risk-Aware Deep RL for Dynamic Portfolio Optimization	arXiv	2025
17	Gaspar & Oliveira	Robo Advising and Investor Profiling	MDPI	2024
18	Shetty et al.	Robo-advisors in financial services	Science Direct	2026
19	Kofman	Scoring the Ethics of AI Robo-Advice	J. of Business Ethics	2025
20	Lamukanyani et al.	Sentiment-Aware Portfolio Optimization (CVaR-Based)	IEEE Access	2025
21	Ahn & Kang	Simul-RL Portfolio Framework: BSM and RL	IEEE Access	2025
22	Pereguda et al.	Impact of the Digital Economy on Investment Portfolio Mgmt	Eastern Euro. Union	2024
23	Kulkarni et al.	Role of robo-advisors in behavioural finance	Cogent OA	2020
24	Enkhsaikhan & Jo	Uncertainty-Aware RL for Portfolio Optimization	IEEE Access	2024

Literature Review (1/24): Regret-Optimized Portfolio Enhancement through DRL and Future Looking Rewards

2025 – Base Paper

- **Relation:** This is the base paper – our project directly adapts its regret-optimized, future-looking reward formulation for goal-based, horizon-sensitive asset allocation.
- **Methodology:** PPO-based DRL framework for dynamic portfolio rebalancing using a regret-based reward (relative to a 60/40 benchmark) combined with a future-looking (forward-horizon) reward component; evaluated via Sharpe ratio and maximum drawdown.

Literature Review (2/24): AI-driven robo-advisors on personal financial management and investment strategies

2024

- **Relation:** Establishes the general case for AI-driven personalized investment advice that our app builds on, but uses no RL-based reward design.
- **Methodology:** Conceptual/survey-style analysis of how robo-advisors use algorithms and ML on market and client data to create tailored, automated investment plans.

Literature Review (3/24): AI Revolution in Wealth Management: Applications and Horizons

2025

- **Relation:** Provides industry-level context on AI adoption in wealth management, motivating the case for an AI-driven planning app, though not a DRL-based technical contribution.
- **Methodology:** Industry-perspective review discussing AI applications (compliance automation, personalization, back-office automation) in wealth management.

Literature Review (4/24): A Multifaceted Approach to Stock Market Trading Using Reinforcement Learning

2024

- **Relation:** Shares the RL-for-finance foundation with the base paper, but focuses on multi-stock trading rather than goal-based portfolio allocation.
- **Methodology:** Uses Advantage Actor-Critic (A2C) and Deep Deterministic Policy Gradient (DDPG) with a novel risk-aware reward (PSR) combining technical and fundamental indicators; validated on 30 Dow Jones stocks.

Literature Review (5/24): A Novel RMS-Driven Deep RL for Optimized Portfolio Management in Stock Trading

2025

- **Relation:** Closely related to the base paper's risk-aware reward design philosophy – both incorporate drawdown-based rewards, though this paper adds sentiment rather than regret/future-looking terms.
- **Methodology:** Proposes PPO_RMS, A2C_RMS, and DDPG_RMS models combining earnings-report data, Max Drawdown-based rewards, and sentiment indicators (RMS); DDPG_RMS outperforms the DJI benchmark.

Literature Review (6/24): Complex forecasting and investment strategy optimization via chain-of-thought of LLMs

2025

- **Relation:** Explores an alternative (LLM-based) path to investment decision-making, contrasting with the base paper's RL-based approach; useful for comparing reasoning-driven vs. reward-driven strategies.
- **Methodology:** Applies chain-of-thought (CoT) prompting with ChatGPT-4.0 on anonymized financial statements of 608 Chinese A-share firms (2010–2023) to predict earnings direction and backtest a trading strategy.

Literature Review (7/24): Detecting and adapting to crisis pattern with context based DRL

2020

- **Relation:** An alternative robustness strategy to the base paper – uses contextual macro features instead of regret/future-looking reward shaping to handle adverse market regimes.
- **Methodology:** Two sub-network DRL architecture; one network processes past strategy performance/volatility, the other processes macro-context features (risk aversion, economic surprise index, cross-asset correlation), with adversarial training for robustness.

Literature Review (8/24): Effectiveness of SIP Investment Among Retail Investors in Jharkhand

2024

- **Relation:** Grounds our project's Indian retail-investor context and validates SIP as the primary savings vehicle our app should recommend for goal-based plans.
- **Methodology:** Primary survey-based study assessing awareness, adoption, and perceived effectiveness of Systematic Investment Plans (SIPs) among retail investors in the Jharkhand region.

Literature Review (9/24): Exploring the potential of AI in personal financial planning and advisory services

2024

- **Relation:** Supports the motivation for our app's AI-driven advisory features (automation, predictive analytics, risk profiling), though it is a conceptual/industry piece rather than a DRL contribution.
- **Methodology:** Discussion-style analysis of AI use cases (automation, predictive analytics, risk assessment) in personal financial planning and advisory workflows.

Literature Review (10/24): Forecasting inflation using disaggregates and machine learning

2023

- **Relation:** Relevant to our goal-cost projection module – our app must inflate future goal costs (e.g. education, car price) accurately, and this paper's ML forecasting approach can inform that sub-module.
- **Methodology:** Aggregates disaggregated (bottom-up) inflation forecasts using linear and nonlinear ML models, applied to Brazilian inflation data, and compares against direct aggregate forecasting.

Literature Review (11/24): From Time Series to Images: A GAF-ResNet-LSTM Framework for Portfolio Optimization

2025

- **Relation:** An alternative deep-learning approach (image-based) to asset selection that could complement our recommendation engine's fund-selection stage, contrasting with the base paper's pure RL approach.
- **Methodology:** Converts price sequences into Gramian Angular Field (GAF) images, extracts spatial features via ResNet and temporal features via LSTM, then applies nested clustered optimization (NCO); validated on NIFTY 50 constituents.

Literature Review (12/24): Hybrid LSTM and PPO Networks for Dynamic Portfolio Optimization

2025

- **Relation:** Directly extends the base paper's PPO approach with a forecasting layer – a natural reference for a possible future enhancement to our recommendation engine.
- **Methodology:** Combines an LSTM forecasting sub-module with a PPO-based allocation agent so decisions are informed by learned temporal price patterns.

Literature Review (13/24): Machine learning in budget forecasting for corporate finance

2024

- **Relation:** While corporate-focused, the proposed ML-driven forecasting architecture is conceptually transferable to our app's personal budget/savings-capacity forecasting module.
- **Methodology:** Conceptual model integrating regression, time-series analysis, and neural networks into a budget forecasting architecture for corporate financial planning.

Literature Review (14/24): Parallel and Multi-Stage Knowledge Graph Retrieval for Financial Asset Recommendations

2025

- **Relation:** Closely relevant to our app's recommendation engine design – shows how to align recommendations with investor behavior, complementing the base paper's reward-based personalization.
- **Methodology:** Knowledge-graph-based retrieval system evaluated on the FAR-Trans dataset, using strict temporal cutoffs to avoid data leakage when generating behaviorally-aligned asset recommendations.

Literature Review (15/24): Regret-Optimized Portfolio Enhancement through DRL and Future Looking Rewards

2025 – Base Paper

- **Relation:** This is the base paper – our project directly adapts its regret-optimized, future-looking reward formulation for goal-based, horizon-sensitive asset allocation.
- **Methodology:** PPO-based DRL framework for dynamic portfolio rebalancing using a regret-based reward (relative to a 60/40 benchmark) combined with a future-looking (forward-horizon) reward component; evaluated via Sharpe ratio and maximum drawdown.

Literature Review (16/24): Risk-Aware Deep Reinforcement Learning for Dynamic Portfolio Optimization

2025

- **Relation:** Closest methodological sibling to the base paper – both use PPO with risk-sensitive rewards, useful for comparing regret-based vs. Sharpe-based reward design in our engine.
- **Methodology:** Integrates a Sharpe-ratio-based reward with direct risk controls (maximum drawdown, volatility constraints), trained using PPO; benchmarked against mean-variance and equal-weighted portfolios.

Literature Review (17/24): Robo Advising and Investor Profiling

2024

- **Relation:** Directly relevant to our risk-profiling module – proposes an objective, quantifiable risk measure that our app's questionnaire-based profiling could adopt instead of vague risk buckets.
- **Methodology:** Introduces an investor profiling method based solely on Relative Risk Aversion (RRA), deriving optimal allocations and comparing them against real robo-advisor (Riskalyze) portfolios for conservative, moderate, and aggressive profiles.

Literature Review (18/24): Robo-advisors in financial services: Redefining wealth management in the age of AI

2026

- **Relation:** Provides a structured typology of robo-advisor institutional forms, useful for positioning our app's design choices within the broader robo-advisory landscape.
- **Methodology:** Conceptual model (grounded in Dubin's theory-building framework) categorizing robo-advisors into Wealth Management Firm, Regulatory, and Personal types, linking institutional design to return maximization, risk minimization, and financial inclusion outcomes.

Literature Review (19/24): Scoring the Ethics of AI Robo-Advice: Why We Need Gateways and Ratings

2025

- **Relation:** Important for our project's ethical/regulatory considerations – highlights trust and transparency issues our app must address when recommending investment allocations.
- **Methodology:** Proposes a regulatory framework requiring robo-advisors to clear four licensing gateways for an AI Robo-Advice License (AIRAL), followed by ongoing ethical-compliance monitoring via a balanced scorecard rating.

Literature Review (20/24): Sentiment-Aware Portfolio Optimization: CVaR-Based Diversification With DRL

2025

- **Relation:** Shares the risk-control philosophy of the base paper (CVaR vs. regret) and shows how sentiment can be incorporated into the state space, a possible future extension for our engine.
- **Methodology:** Uses DDPG to dynamically adjust allocations under Conditional Value-at-Risk (CVaR) constraints while incorporating FinBERT-derived sentiment scores into the state representation, accounting for transaction costs.

Literature Review (21/24): Simul-RL Portfolio Framework: Black-Scholes-Merton and RL for Asset Allocation

2025

- **Relation:** Addresses a limitation relevant to our project – limited historical data for personal goal scenarios – by showing how simulated data can augment RL training, unlike the base paper which relies on historical market data alone.
- **Methodology:** Uses Monte Carlo simulation data (grounded in the Black-Scholes-Merton model) to increase training data complexity and diversity for an RL-based asset allocation agent, addressing overfitting and limited statistical structure in real data.

Literature Review (22/24): The Impact of the Digital Economy on Investment Portfolio Management

2024/25*

- **Relation:** Provides macro-level context on how digital transformation is reshaping portfolio management practices, motivating our project's digital-first (mobile app) approach.
- **Methodology:** Empirical/analytical study examining how digital economy indicators relate to investment portfolio management and investor behavior in stock markets.

Literature Review (23/24): The role of robo-advisors in behavioural finance, shaping investment decisions

2020

- **Relation:** Relevant to our app's UX and risk-communication design – highlights behavioral biases robo-advisors must account for, complementing the base paper's purely quantitative reward design.
- **Methodology:** Longitudinal case study examining robo-advisors through the lens of behavioural finance and their role in shaping investors' financial decision-making processes.

Literature Review (24/24): Uncertainty-Aware Reinforcement Learning for Portfolio Optimization

2024

- **Relation:** Extends the base paper's risk-awareness with explicit uncertainty quantification (Bayesian forecasting), a potential enhancement for our engine's confidence-aware recommendations.
- **Methodology:** Proposes the Uncertainty-Aware Markov Decision Process (UAMDP), coupling Bayesian forecasting, posterior-sampling RL (Thompson sampling), and CVaR-constrained planning; improves Sharpe ratio and roughly halves maximum drawdown vs. baselines.

- **Robo-advisory and AI financial planning literature** covers general AI-driven advisory concepts, investor profiling, ethics, and Indian SIP adoption – but none integrate a *dynamic, reward-shaped* recommendation engine tied to a specific goal's remaining time horizon.
- **No existing system combines:**
 - ① Goal-based horizon-aware planning (amount, deadline, inflation-adjusted target),
 - ② A mobile-first, India-relevant recommendation engine spanning mutual funds, equity, and debt instruments.
- **This project's contribution:** bridging goal-based financial planning with regret-based, future-looking DRL reward shaping in a single mobile recommendation system.

Project Planning: Project Calendar

Phases	July 2026	August 2026	September 2026	October 2026	November 2026	December 2026	January 2026	February 2026
Requirement Analysis and Literature Review								
System Design								
Frontend Development								
Backend Development								
Recommendation Engine Development								
Integrating Modules								
Testing and Bug Fixes								
Deployment and Documentation								

Project Planning: Phase-wise Timeline

- **July 1 – August 10:** Requirement analysis, literature review finalization
- **August 11 – September 5:** System design – UML diagrams, architecture
- **September 6 – November 15:** Frontend Development
- **September 6 – November 15:** Backend development Development
- **September 6 – November 25:** Recommendation engine development & training
- **November 26 – December 25:** Integration of frontend, backend, and recommendation engine
- **December 21 – February 5:** Testing, bug fixing, and evaluation against benchmarks
- **January 26 – February 28:** Documentation and final presentation preparation

Primary:

- UI/UX architecture
- Mobile screens
- Goal creation interface
- Dashboard
- Recommendation visualization

Supporting:

- Frontend-backend integration
- User testing

Primary:

- Risk questionnaire
- Risk tolerance
- Risk capacity
- Risk classification
- User financial profile

Supporting:

- Risk-related literature analysis
- Testing different user profiles

Amarthya Regimon Joseph — AI Recommendation & Portfolio Allocation

Primary:

- Recommendation engine
- Time-horizon analysis
- Asset allocation
- Dynamic portfolio adjustment
- Base paper's DRL concepts

Supporting:

- PPO/reward mechanism study
- Benchmark comparison
- Recommendation evaluation

Primary:

- Inflation-adjusted goal amount
- Future Value calculation
- SIP calculation
- Expected investment growth
- Goal feasibility
- Financial/market data processing

Supporting:

- Forecasting model/data analysis
- Financial API integration
- Validation of financial calculations

System Architecture: Level 0 (Context Diagram)

level_0.png

System Architecture: Level 1

level_1.png

System Architecture: Level 2

level_2.png

Modular Breakdown

Module 1: Mobile Application & UI/UX

- UI/UX architecture, Mobile Screens, Recommendation visualization.
- Goal Creation Interface, Dashboard.

Module 2: AI Recommendation and Portfolio Allocation

- Recommendation Engine, Time Horizon Analysis.
- Asset Allocation, Portfolio Reallocation.

Module 3: Risk Profiling and & Modelling

- Risk Questionnaire, Risk Tolerance, Risk Capacity.
- Risk Classification, User financial Profile.

Module 4: Financial Forecasting & Financial Computation

- Inflation adjusted Goal amount, SIP Calculation.
- Expected Investment Growth, Market Data Processing.

Tools and Technologies Used

- **Frontend (Mobile App):** Flutter
- **Backend:** Python (FastAPI)
- **Database:** Supabase (Postgres)
- **Recommendation Engine:** Python, PyTorch / Stable-Baselines3
- **Version Control:** Git & GitHub

Functional Requirements

- **Financial Goal Management:** Allow users to define financial goals with target amount and time horizon.
- **Risk Profiling:** Assess the user's risk tolerance and risk capacity to classify the user's risk profile.
- **Financial Calculation:** Calculate inflation-adjusted goal amount, required SIP contribution, and expected investment growth.
- **Investment Recommendation:** Recommend suitable allocation across mutual funds, equity, and debt instruments based on the user's goal and risk profile.
- **Dynamic Portfolio Adjustment:** Adjust asset allocation based on the remaining goal horizon, risk profile, and changing market conditions.

Non-Functional Requirements

- **Usability:** The application should provide a simple and intuitive mobile-first interface.
- **Performance:** Financial calculations and investment recommendations should be generated efficiently.
- **Security:** User personal and financial information should be securely handled and stored.
- **Reliability:** Financial calculations and recommendation results should be consistent and dependable.
- **Accuracy:** Goal calculations, SIP calculations, inflation adjustments, and portfolio recommendations should provide reliable results.

Conclusion

- Existing goal-based financial planning tools rely on static, rule-based risk assessment that does not adapt to a goal's remaining time horizon or evolving market conditions.
- The reviewed literature demonstrates strong individual advances in robo-advisory design, and behavioral/ethical considerations – but no single system unifies these for **personal, goal-based** investment planning.
- This project proposes a mobile application that adapts the base paper's regret-optimized, future-looking reward mechanism to recommend dynamic investment allocations for individual financial goals.
- This phase 1 review establishes the problem statement, objectives, literature foundation, system design, Modular division and Technological requirements.

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Thank You